

Design Project of a 25 W Class AB Power Amp

With 4 Ohm Resistive Load

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Abstract—This study presents the design, simulation, and hardware implementation of a discrete-component Class AB audio power amplifier intended for high-fidelity applications. The amplifier is structured to replicate the internal architecture of a conventional operational amplifier while overcoming the power and thermal limitations of integrated circuit implementations. Using a backward, specification-driven analytical design approach that begins from the given load and output power requirements, the circuit was developed to deliver a target output power of 25 W into a 4 Ω resistive load while operating from a ± 22 V dual-rail power supply and maintaining a usable audio bandwidth of 20 Hz to 20 kHz. The design employs a three-stage topology consisting of a differential input stage, a high-gain voltage amplification stage, and an improved push-pull Class AB output stage using power-rated bipolar junction transistors.

Keywords—Audio power amplifier, Class AB amplifier, discrete operational amplifier, push-pull output stage

I. INTRODUCTION

Audio power amplifiers are one of the most important components in sound system applications, as they serve as the bridge between low-level signal sources and loudspeakers, requiring both linearity and signal driving capability. For this design project, the circuit is structured as an implementation of a high-power Operational Amplifier (Op-Amp). Conventional IC op-amps are widely used in signal-processing applications, but their power-handling capability and thermal performance are inherently limited. To address these constraints, the design reconstructs the essential internal architecture of a standard op-amp using power-rated transistor devices. The discrete amplifier retains the traditional three-stage topology. The first

stage utilizes a differential input pair to manage error sensing and negative feedback, which drives the second stage, which is a high-gain Voltage Amplification Stage for primary voltage gain. And lastly, the third stage, which is a Class-AB output stage designed to deliver the necessary load current.

This design choice enables the system to utilize the high open-loop gain and low distortion characteristics inherent to op-amps, while also providing the ruggedness necessary to drive a low-impedance load. The primary objective is to deliver 25 Watts into a 4-ohm resistive load with high fidelity. By constructing the op-amp using discrete parts (such as TIP3055/2955 transistors), the design achieves the power targets that a standard Integrated Circuit (IC) cannot, while satisfying the requirement for a discrete circuit implementation.

II. IMPLEMENTATION

A. Circuit Design

The circuit design was carried out using an analytical approach, starting from the required load and output power specifications. Since the amplifier is intended to deliver 25 W into a 4 Ω resistive load, the necessary output voltage swing and peak load current were first determined. These values serve as the basis for selecting the power supply rails, transistor current and voltage ratings, and the power ratings of passive components.

All component values and biasing conditions were chosen to ensure linear operation across the full

signal range while keeping each device within its safe operating limits. Conservative assumptions were made for transistor parameters, such as base-emitter voltage drops and current gain at high collector currents, to account for worst-case operating conditions. In addition, sufficient voltage and current headroom were provided to prevent output clipping and excessive thermal stress during continuous operation.

This design approach ensures that the amplifier can meet the target output power reliably, while maintaining stability and safe operation. The detailed calculations for each stage of the circuit, including the power supply, output stage, driver stage, bias network, voltage amplification stage, and input stage, are presented in the following subsections.

B. Power Supply Calculation

The first backward solution involves calculating the required voltage swing to support the required 25W power target. Using Ohm's Law for power, the necessary RMS voltage is determined by:

$$V_{RMS} = \sqrt{P_{Target} \times R_{Load}} = \sqrt{25W \times 4\Omega} = 10V_{RMS}$$

The amplifier must reproduce the full AC waveform without introducing clipping, hence, the peak voltage can be calculated from the RMS value.

$$V_{Peak} = V_{RMS} \times \sqrt{2} = 10V_{RMS} \times 1.414 = 14.14V$$

Based on this, a standard supply of 22V to provide headroom for the circuit.

C. Output Stage (Q8, Q9, R14. and R15)

The peak current required by the 4 Ω load will determine the rating of the output transistors.

$$I_{Peak} = \frac{V_{Peak}}{R_{Load}} = \frac{14.14V}{4\Omega} = 3.54A$$

Based on the result of the calculation. The TIP3055 (NPN) and TIP2955 (PNP). These transistors having a rating of 15A / 60 V can handle the required 3.54A. In this amplifier circuit, the R14 and R15 provide thermal stability and current sharing. Which is why they will need a sufficient power rating. The voltage drop at peak current was calculated to ensure stability without excessive power waste.

$$V_{Drop} = I_{Peak} \times R = 3.54A \times 0.33\Omega = 1.16V$$

Using the peak current calculated (3.54A), the peak instantaneous power dissipation (P_{Peak}) across the resistor is calculated as:

$$P_{Peak} = I_{Peak}^2 \times R = (3.54A)^2 \times 0.33\Omega$$

$$P_{Peak} = 12.53 \times 0.33 = 4.13W$$

To provide headroom, a 5W resistor provides a safe margin above the 4.13 W peak, this will ensure that the component will remain at safe temperature.

D. Driver Stage

The driver stage is necessary because the chosen output transistors (TIP3055) exhibit low current gain (β) when operating at high currents. To address this, a "Driver" is utilized to amplify the current from the voltage stage, effectively forming Darlington configurations with the Q13/Q9 and Q12/Q8 pairs. Assuming a worst-case β of 20 at 4A for the TIP3055, the driver is required to provide the following base current:

$$I_{Base} = \frac{3.54A}{20} = 177mA$$

To meet this requirement, the transistors TIP41C and TIP42 were selected; since as 6A devices, they can handle the 177mA requirement without issue. Additionally, 100 Ω valued resistors (R_{11} and R_{13}) were included to prevent parasitic oscillation.

E. Bias Generator (Q2)

For the amplifier to operate as Class AB and eliminate crossover distortion, the bases of the driver transistors must be separated by a specific voltage. The output of this amplifier utilizes a Darlington configuration (Driver + Output), therefore, the bias generator must overcome four base-emitter junctions (two NPN and two PNP). The total bias voltage required is:

$$V_{Bias} = 4 \times 0.65V = 2.6V$$

F. Voltage Amplification Stage

The Voltage Amplification Stage provides the voltage gain of the amplifier. This stage must run sufficient bias current from around 6 mA to 10 mA in order to drive the capacitance of the output stage. For this stage, it is bootstrapped using Resistors R_8 and R_9 to simulate a constant current source which will increase gain. The current is calculated as:

$$I_{VAS} \approx \frac{V_{CC}}{R_8 + R_9} \approx \frac{22V}{2.4k\Omega} \approx 9mA$$

This 9mA bias current will ensure speed. Furthermore, emitter degeneration resistor R_5 is used to stabilize the DC operating point. The voltage drop across it is calculated to verify linearity.

$$V_{Degeneration} = 9mA \times 68\Omega \approx 0.6V$$

G. Input Differential Pair (Q14, Q16)

The input stage compares the incoming signal to the feedback signal. The total tail current flowing through resistor R_1 is critical for setting the operating point. With a supply voltage of approximately 22V, the voltage across R_1 is derived as:

$$V_{R1} = V_{CC} - 0.7V$$

$$V_{R1} = 22 - 0.7V$$

$$V_{R1} = 21.3V$$

The tail current is calculated as:

$$I_{TAIL} = \frac{21.3V}{R_1} = \frac{21.3V}{4.7k\Omega} \approx 4.5mA$$

This current is split between transistors Q_{14} and Q_{16} (2.25mA each). While this is considered high current, it is acceptable for achieving lower noise. Finally, the input impedance resistor R_2 serves to minimize DC offset voltage. To achieve this, R_2 must match the feedback resistor R_f . Since R_f is 22k Ω , R_2 is correctly set to 22k Ω .

H. Feedback Network and Sensitivity (R_f, R_{22})

The feedback network determines the overall closed-loop gain and input sensitivity of the amplifier. The gain factor is established by the ratio of the feedback resistor to the shunt resistor. Using the selected component values ($R_f = 22k$, $R_{22} = 1.5k$), the gain is calculated:

$$Gain = 1 + \frac{R_f}{R_{22}} = 1 + \frac{22000}{1500} \approx 15.6$$

This gain dictates the input voltage required to drive the amplifier to its full 25W output. To reach the calculated peak output voltage of 14.14V, the required input sensitivity is derived as :

$$V_{IN} = \frac{V_{PEAK}}{GAIN} = \frac{14.14}{15.6} \approx 0.9V_{PEAK} \approx 0.64V_{RMS}$$

I. Passive Filters and Bandwidth Control

To ensure that the amplifier has stable operation and handles audio frequencies correctly, three specific passive networks are integrated into the design. Input High Pass Filter (C_1, R_2) The input coupling capacitor (C_1) forms a high-pass filter with the input impedance resistor (R_2) to prevent stray DC voltage from the source from entering the amplifier. The cutoff frequency (f_c) was calculated to ensure it remains well below the audible range (20Hz). Using the selected values of $R_2 = 22k\Omega$ and $C_1 = 47\mu F$:

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 22000\Omega \times 47 \times 10^{-6}F} \approx 0.15Hz$$

This result of 0.15Hz confirms that the filter allows all deep bass frequencies to pass through without attenuation.

J. Feedback DC Blocking Filter.

To minimize DC offset at the speaker output, capacitor C_2 is placed in series with the feedback shunt resistor R_{22} . This lowers the amplifier's gain to unity (1) at DC (0 Hz) while maintaining full gain at audio frequencies. The cutoff frequency for this network is calculated as follows :

$$f_c = \frac{1}{2\pi R_{22}C_2} = \frac{1}{2\pi \times 1500\Omega \times 100 \times 10^{-6}F} \approx 1.06Hz$$

The calculated cutoff of approximately 1Hz ensures excellent low-frequency response while maintaining DC stability.

K. Design Considerations

For the push pull output stage, as the current swing goes up to 3.54A and the voltage swing goes up to 14V, a TIP3055 (NPN) and TIP2955 (PNP) was used. Both transistors can handle collector currents up to 15A and voltages up to 60V according to its datasheet. Furthermore, as these transistors dissipate the most power other than the load, these transistors must be able to handle high temperatures. Both TIP3055 and TIP2955 are able to have a maximum operating temperature of 150 degrees celsius. Furthermore, these transistors come in a TO-247 package which makes heat sink mounting easier.

As this design utilizes a darlington pair to drive the output stage transistors, appropriate current driving

transistors must be used. Hence a pair of TIP41C and TIP42C was used. These transistors will be able to boost the current gain while maintaining stability as these transistors are able to handle up to 100V.

In order to set the bias voltage between the bases of the driver transistors, a transistor as a diode was used. Using a transistor instead of a diode provides better thermal stability and linearity. As a diode heats up, its bias voltage will also change which could affect the output stages. Furthermore, this transistor could be used as a V_{be} multiplier which can be used to fine tune the biasing base voltage.

The input stage utilizes the 2N5401 PNP transistor to form the differential pair. The 2N5401 was specifically chosen for its high voltage breakdown rating and low noise characteristics. As this is the very first stage of the amplifier, any noise introduced here would be amplified by following stages. Additionally, its high voltage rating provides a significant safety margin against power supply spikes, ensuring the sensitive input stage remains stable well within its safe operating area.

Lastly, for the Voltage Amplification Stage, the 2N5551 NPN transistor is used as it is the NPN counterpart of the 2N5401 transistor which also has high voltage rating and sufficient gain.

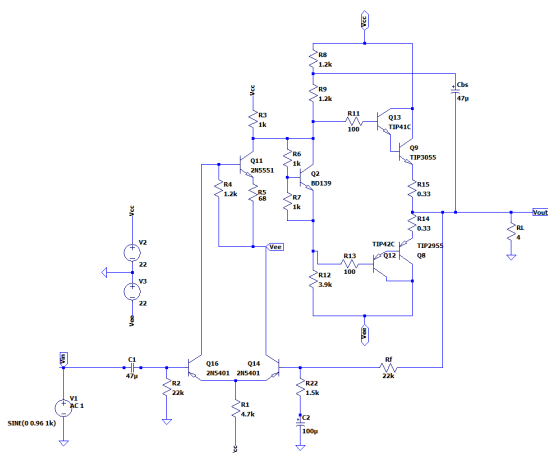


Figure 1.0 - Final Circuit Design using LTSpice

L. PCB Design and Layout

After the LTSpice Simulations and Design Considerations, the group proceeded with constructing the prototype of the PCB using the software Icad. In this case, the initial process is the familiarization with the software. The group proceeded with the initial drafts using the Icad software following the design of the final circuit design from the LTSpice simulation.

M. Final PCB Layout

After constructing the first prototype using the software, the initial observations are as follows: First the transistors should be placed at the edge of the PCB. Second, the wires used should be thick enough so that they will not act as a resistor due to the high currents that flow in the circuit. For the traces, a 2.3mm thickness was used in order to provide sufficient space for the current. Furthermore a ground plane was introduced on the empty spaces in order to improve signal integrity and reduce impedance. The third realization is that a volume control is needed, specifically a 50k Ω potentiometer; an aux port is also needed for the input, speaker output terminals and input biasing terminals are also needed to complete the PCB Design

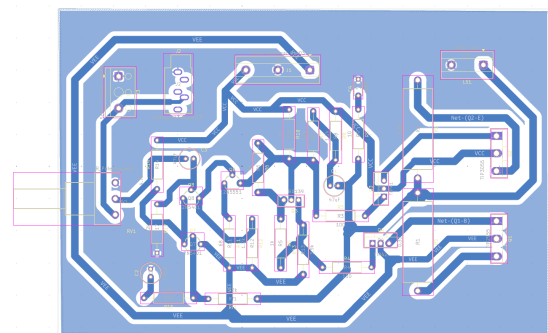


Figure 2.0 - Final PCB Layout

The figure shown shows the final layout of the PCB designs which has improved design choices. The Final PCB Design is shown in Figure 2.0, this design will be printed using the etching method.

N. PCB Prototype Fabrication

Before the actual etching process of the PCB, the group initially prepared the materials in the design. Firstly, the group bought the Photosensitive PCB. Additionally, a printed layout of the PCB was prepared for the photolithography process of the PCB etching. The PCB fabrication process begins by defining the core circuit pattern on the copper surface through photolithography. This process starts with coating the PCB with a photoresist, which is a light-sensitive polymer layer. The circuit design which was printed on a coupon bond paper, is then precisely aligned over the coated board. The transfer of the pattern occurs during exposure to strong UV light, which causes the photoresist to selectively harden or become soluble, depending on its chemical type.

After the exposure to light, the next step removes the soluble portions of the photoresist using a

sodium hydroxide solution, revealing the underlying copper in the exact form of the circuit pattern. The unwanted exposed copper is subsequently removed during the Etching phase using ferric chloride. Lastly, the remaining photoresist is completely stripped away, leaving only the intended copper traces that form the functional circuitry of the PCB.

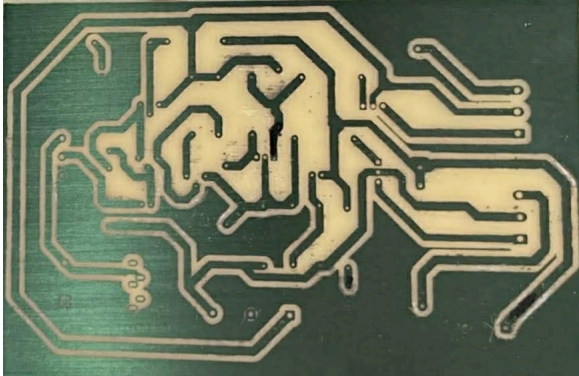


Figure 3.0 - Fabricated PCB for the Power Amplifier Circuit

O. PCB Assembly

After completing the PCB fabrication process and having the finished copper traces, the next step was to assemble the electronic components onto the fabricated board. The PCB can then be converted into a fully functional circuit. The components, including transistors, resistors, capacitors, the 50 k Ω potentiometer, and the input and output terminals, were placed on the board according to the finalized PCB layout. To ensure full functionality, proper orientation and correct placement were observed, particularly for polarized components. Once all components were mounted correctly, soldering was performed to create and ensure reliable electrical connections between the components and the copper traces. The goal of this step is to ensure clean solder joints and to avoid solder bridges that could lead to short circuits. After soldering, the PCB was visually inspected to verify that all connections were secure and consistent with the intended design.



Figure 4.0 - Final Power Amplifier Prototype

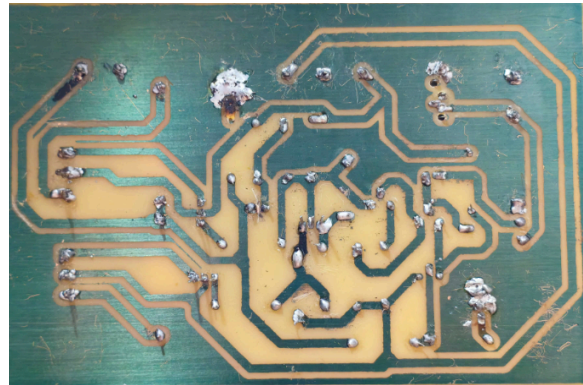


Figure 4.1 - Final Power Amplifier Prototype

P. Circuit Testing and Verification

After the assembly of the circuit, the initial testing was done using a multimeter to check continuity and detect possible short circuits. This ensured that the power, ground, and signal paths were correctly connected. Once these checks were completed, the circuit was powered on under controlled conditions using a DC input of 22 V, 4 A, which was based on the LTSpice simulation for both V_{CC} and V_{CE} . An input signal from a cellphone was applied through the auxiliary input, and the output was tested using the required 4 Ω speaker load at the speaker terminals to confirm proper operation of the amplifier and volume control. Any issues encountered during testing were addressed through basic troubleshooting and corrective soldering. Successful operation of the circuit confirmed that the PCB design, fabrication, and assembly processes were properly executed, resulting in a functional PCB prototype.

Model Name	Quantity
TIP3055 NPN	1
TIP2955 PNP	1
TIP41C NPN	1
BD139	1
2N5551	1
2N5401	2

Table 1.0 - List of Transistor Used

Value	Rating	Quantity
0.33	5W	2
4.7k	1/4W	1
3.9k	1/2W	1
22k	1/4W	2
1.2k	1W	3
1k	2W	3
100	1/2W	2
68	1/4W	1
1.5k	1W	1

Table 2.0 - List of Resistor Used

Value	Rating	Quantity
47uF	50V	2
100uF	50V	1

Table 3.0 - List of Capacitors Used

III. Design Verification and Results

To verify the amplifier's overall performance, the group emulated the circuit using *LTspice* with the appropriate values of each component computed from the previous sections of this paper. By utilizing the *.tran* SPICE directives, the group performed the transient analysis to measure its output waveform and to plot its collector current waveform, which aids in identifying the class in which the amplifier belongs.

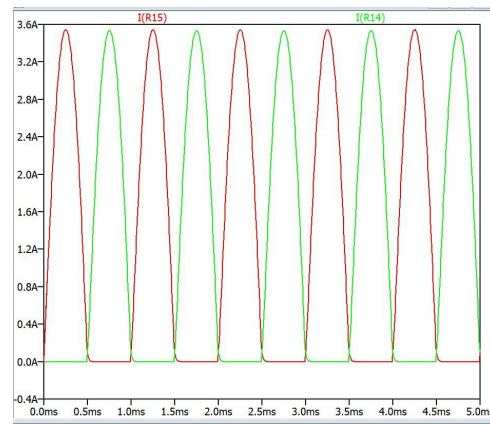


Figure 5.0 Collector Current Waveform

Observing the output waveform, the group immediately noticed a similar pattern in the collector current waveform, with an improved Push-Pull stage amplifier. Observing closely, each waveform can be seen to conduct for slightly more than half a cycle (more than 180 degrees), which further verifies that the class to which the designed amplifier belongs is in AB, specifically an Improved Push-Pull stage.

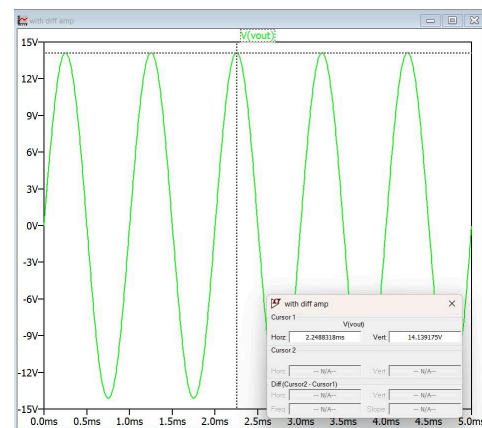


Figure 6.0 Output Voltage Waveform

Moreover, the group was able to obtain the target peak voltage, in which the theoretical value was computed at the methodology part. Having a theoretical peak voltage value of 14.14V, the group measured a value of 14.1392V, in which the percent error was measured:

$$\%error = \frac{|Measured\ Value - True\ Value|}{True\ Value} \times 100\%$$

$$\%error = \frac{|14.1392V - 14.14V|}{14.14V} \times 100\%$$

$$\%error = 0.005\%$$

Observing the percentage error computed in this equation, the group observed an insignificant difference between the measured value and the true value (from computation), thus achieving a highly accurate design.

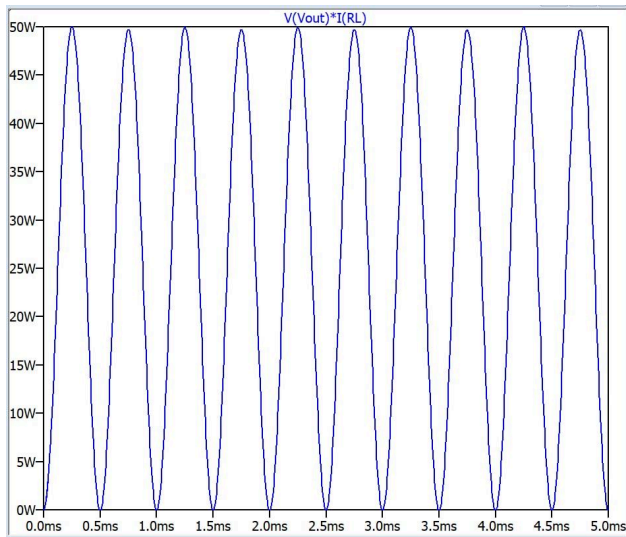


Figure 7.0 Output Power Waveform

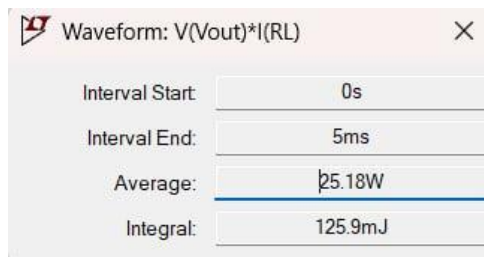


Figure 7.1 Measured Average Power

Furthermore, the group also ensures that the required Power value of 25W is obtained from the design. With that, the group determines the output power by measuring the average value of the Power, which is calculated from the voltage across the output area and the current through the load resistor of the circuit. With that, the obtained average power value is said to be 25.18W. To further measure the accuracy of the design to its target value, the group measured its percent error using the formula:

$$\%error = \frac{|Measured\ Value - True\ Value|}{True\ Value} \times 100\%$$

$$\%error = \frac{|25.18W - 25W|}{25W} \times 100\%$$

$$\%error = 0.72\%$$

Having a percentage error of less than 1 percent indicates high accuracy towards the target theoretical value of the average power; the assumptions are valid, and measurement imperfections or component tolerances have had minimal impact.

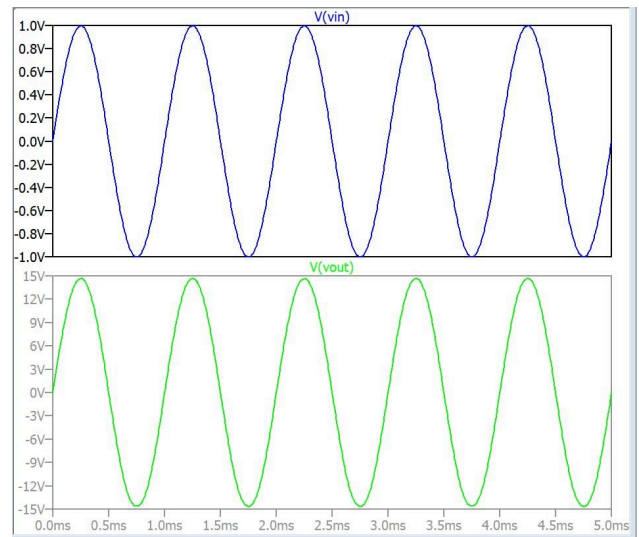


Figure 8.0 Waveform Comparison Between the Input (Blue) and Output (Green) Voltage

Upon observing the waveforms of the input and output voltage, the group can clearly see a significant increase in value as the peak voltage has amplified from 1.0V (in the input waveform) to 15V (in the output waveform). By having an approximate 1:15 ratio, the group can safely conclude that the circuit has provided a good voltage amplification.

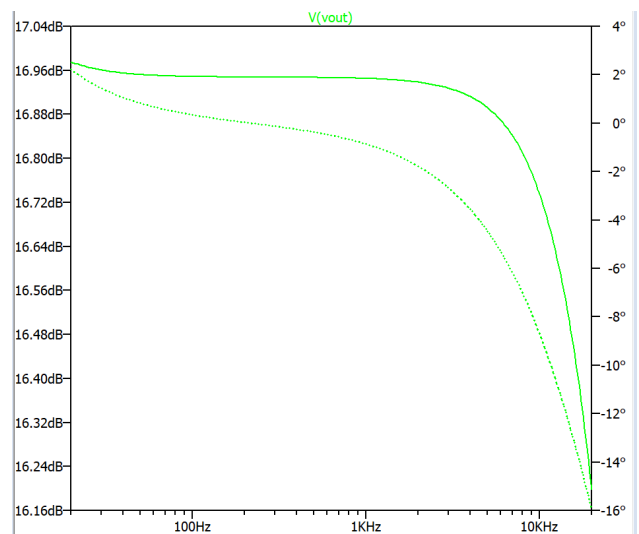


Figure 9.0 Frequency Response of the Output Voltage

From the graph, the group was able to visualize the bandwidth at which the amplification is present. Moreover, the group also observed an almost flat response with minimal roll-off rate in the middle, thus ensuring that the output is linear.

IV. CONCLUSION

In this project design, the group applied the theory discussed in the lecture and designed a power amplifier with the specifications provided by the instructors. Using different variations of circuits, the group initially had a 4Ω resistive load and aimed to amplify the input with a target power value of 25W. After conducting thorough design layout and component calculations, the group was able to design a Class AB amplifier that replicates the behavior of an improved push-pull amplifier. The final design incorporated a three-stage operational amplifier topology consisting of a differential input stage, a voltage amplification stage, and a Class AB push-pull output stage, enabling high linearity, sufficient voltage gain, and adequate current-driving capability for low-impedance loads. By measuring the output terminal of the circuit, the group successfully amplified the input and achieved the target value of 25W with the provided 4Ω resistive load. Simulation and measurement results showed close agreement with theoretical calculations, with the output peak voltage and average output power exhibiting less than one percent error relative to the target values, thereby validating the analytical design approach and component selection. Additionally, frequency response analysis confirmed stable amplification across the audible range with minimal distortion, while transient waveform observations verified proper Class AB operation through conduction angles exceeding 180 degrees per cycle. In the end, the group has successfully designed a Class AB Power Amplifier with the provided specifications and obtained the target output value of the circuit.

V. REFERENCES

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